

Index Image Method

$$C_x = \sum_i^k \frac{x_i}{k} \quad C_y = \sum_i^k \frac{y_i}{k}$$

(C_x, C_y) : coordinates of centroid

(x_i, y_i) : coordinates of i^{th} white point for $i \in [1, k]$

k : # of white pixels = area of ball

Note: The heuristic method you implement in the step before uses nested for-loops, which is SLOW.

Index image implements the same formula while using linear algebra to speed up computations.

Let's go through a visual example on how this is done on a small binary image.

You're given:

$$k = \text{sum of img} \quad (5 \cdot 255 \text{ in this example})$$



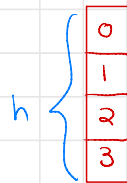
$$\text{img} \in \mathbb{R}^{h \times w}$$

The binary image you created, where 255=white, 0=black



$$x = \text{id}x \in \mathbb{R}^{1 \times w}$$

A vector (transposed) ranging from 0 to (the # of columns of img) - 1



$$y = \text{id}y \in \mathbb{R}^h$$

A vector ranging from 0 to (the # of rows in img) - 1

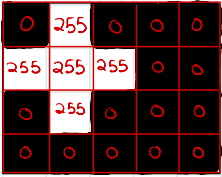
Find C_x :

1. Element-wise multiply img with $x\text{-idx}$ (broadcasting)

$$\hookrightarrow (\text{img} * x\text{-idx})$$

$(h \times w)$ $(1 \times w)$
 $\quad \quad \quad \searrow$
 $\quad \quad \quad (w \times w)$

the dimensions don't match, so Python will duplicate $x\text{-idx}$ into a $h \times w$ matrix



$\text{img} \in \mathbb{R}^{h \times w}$

$\times$

element-wise
 $\downarrow$

0	1	2	3	4
0	1	2	3	4
0	1	2	3	4
0	1	2	3	4
0	1	2	3	4

$\tilde{x}\text{-idx} \in \mathbb{R}^{h \times w}$

$=$

0	1(255)	0	0	0
0	1(255)	2(255)	0	0
0	1(255)	0	0	0
0	0	0	0	0
0	0	0	0	0

$\in \mathbb{R}^{h \times w}$

2. Sum (element-wise) the resulting matrix

$$\hookrightarrow \text{np.sum}(\text{img} * x\text{-idx})$$

$$\text{sum} \left(\begin{bmatrix} 0 & 1(255) & 0 & 0 & 0 \\ 0 & 1(255) & 2(255) & 0 & 0 \\ 0 & 1(255) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \right) = 5 \times 255$$

3. Divide the sum by k

$$\hookrightarrow x = \text{np.sum}(\text{img} * x\text{-idx}) / k \quad \textcircled{1}$$

$$\frac{5 \times 255}{k} = \frac{5 \times 255}{5 \times 255} = 1.0$$

x = This is the x-coordinate of the ball's center

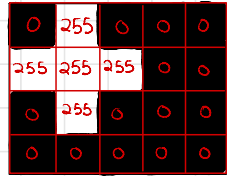
4. Round this result

$$C_x = \text{round}(x)$$

Find C_y

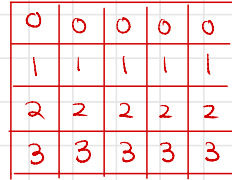
Repeat this process, but with $y\text{-idx}$.

$$\hookrightarrow y = \text{np.sum}(\text{img} * y\text{-idx}) / k \quad (2)$$



0	255	0	0	0
255	255	255	0	0
0	255	0	0	0
0	0	0	0	0
0	0	0	0	0

$\text{img} \in \mathbb{R}^{h \times w}$



0	0	0	0	0
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
3	3	3	3	3

$\times$

$y\text{-idx} \in \mathbb{R}^{h \times w}$

$$C_y = \text{round}(y) = 1$$

Combine into center coordinate

$$\hookrightarrow \text{center} = (\text{round}(x), \text{round}(y)) \quad (3)$$

round since circle() can only take integers

Find radius

use $A = \pi r^2$

(A is pixels squared, r is pixels)

$$A = \frac{k}{255}$$

solve for r

$$r = \sqrt{\frac{k}{255 \cdot \pi}}$$

(remember to round this)

$$\rightarrow \text{radius} = \text{round}(\text{np.sqrt}(k/255/\text{np.pi})) \quad (4)$$

Return values

$$\hookrightarrow \text{return center, radius} \quad (5)$$

TLDR: Use (1), (2), (3), (4), (5) in your code